



GHANA COMMUNICATION TECHNOLOGY UNIVERSITY
FACULTY OF COMPUTING AND INFORMATION SYSTEMS
DEPARTMENT OF COMPUTER SCIENCE

Course:

CSSD 232 - Automata Theory

Focus:

Finite Automaton: Equivalence DFAs from NFAs

Lecturer:

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Date: 14TH July, 2026.

Instructions:

1. After completing each of your task, create a report that includes the following:
 - a) Cover Page
 - b) Images of the transition diagrams.
 - c) Transition tables (where applicable)
2. Ensure to label your machines properly.
3. Student should ensure to submit the report on the LMS.
4. Ensure to take note of the deadline for the submission of the tasks.

Tasks:

1. Construct a DFA and an NFA that both recognize the following languages.
Both machine must recognize the same language for each language provided.

a)

- i. Set of strings that begin with 001
- ii. Set of strings that ends with abc
- iii. Set of strings that contains at least one copy of bc
- iv. Set of strings that has a substring of 010
- v. Set of strings that ends with 12

- b) For the NFA in a(v) above, identify 5 strings that the machine can recognize.
- c) Using the extended transition function, process 3 strings from your answer in b.
- d) Construct the equivalence DFA for the NFA obtained from a.(iv) and a.(ii) above.

2. Construct the equivalence DFA from the transition table provided on **page 45 and 46 in the slide: Finite Automaton.**

3. Given the Epsilon NFA below, perform the epsilon closure for each state in the diagram.

